SOC Incident Report: XSS Attempt Detection

An alert (Event ID: 116) was triggered due to malicious JavaScript code detection within a requested URL, signifying a potential Cross-Site Scripting (XSS) attack.

Investigation Findings:

- External source attempted multiple XSS payload injections against an internal web server.
- Activity confirmed as malicious; however, no evidence of successful exploitation was found.

TRUE POSITIVE

Unsuccessful Attack – No escalation to Tier 2 required.

Incident Analysis: Alert & Network Information

1. Alert Details

Event ID: 116

Alert Name: JavaScript Code Detected in Requested URL

Detection: Signature-Based (Rule: SOC166)

2. Network Forensics

Source IP
`112.85.42.13`

Destination IP
`172.16.17.17`

Port/Protocol
443 (HTTPS)

Method
GET Request

Action
Allowed

Attack Behavior & Lifecycle Analysis

3. Traffic Direction

Internet → Company Network

Public IP targeting private server.
Confirms external attacker intent.

5. HTTP Response Analysis

200 OK: Normal Requests
302 Found: Malicious Redirect

No payload reflection observed.

6. Success Evaluation

UNSUCCESSFUL

Payload Sent: **YES**
Reflected: **NO**
Execution: **NO**

Conclusion: Infrastructure prevented exploitation.

4. Attack Behavior Analysis

Phase 1: Reconnaissance

Accessed normal endpoints to understand structure: `/, /about-us/, /search/?q=test`

Phase 2: Payload Testing

Tested JS reflection via: `?q=prompt(8)`

Phase 3: Exploitation Attempts (XSS Payloads)

1. Image-based XSS

```
?q=<$img src=q onerror=prompt(8)$>
```

2. SVG + Script Injection

```
?q=<$svg><$script ?>$alert(1)
```

3. Obfuscated JS

```
?q=<$script>$for((i)in(self))eval(i)(1)<$/script>
```

4. JS Protocol Injection

```
?q=<$script>javascript:$alert(1)</script>
```

Observation: Multiple variations detected.

SOC Incident Summary

7. Attack Evaluation

Payload Sent: **YES**

Payload Reflected: **NO**

Execution Observed: **NO**

Conclusion: The attack was unsuccessful.

8. Classification

Type
XSS

Detection
True Positive

Outcome
Unsuccessful

Impact
None

9. Escalation Decision

NO ESCALATION

Reason:

- No compromise
- No execution

10. Incident Artifacts

1. Source IP (Attacker)

`112.85.42.13`

External attacker performing XSS attempts

2. Destination IP (Target)

`172.16.17.17`

Internal web server

3. Malicious URL Payload (Example)

```
https://172.16.17.17/search/?q=<$script>javascript:$alert(1)</script>
```

Primary XSS payload detected in request

4. Alternative Image Payload

```
?q=<$img src=q onerror=prompt(8)$>
```

XSS attempt via image onerror event

11. FINAL CONCLUSION

External XSS injection attempt failed due to robust server-side behavior (302 redirects and lack of reflection).

No impact observed; case closed.

Status

CLOSED